

Supplementary Experiments and Extended Results

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This document provides full methods, tables, and discussion for the supplementary experiments referenced in the main text. Experiment numbering follows the pre-registration.

Contents

1	Experiment 1 (H1): Toxicity correction	2
1.1	Hypothesis	2
1.2	Methods	2
1.3	Results	2
1.4	Discussion	2
2	Experiment 2 (H2): Broad-mechanism drugs	3
2.1	Hypothesis	3
2.2	Methods	3
2.3	Results	4
2.4	Discussion	4
3	Experiment 4 (H4): Localization score	5
3.1	Hypothesis	5
3.2	Methods	5
3.3	Results	5
3.4	Discussion	6
4	HP2: Pathway function prediction from corrected Perturb-seq distances	6
4.1	Hypothesis	6
4.2	Methods	6
4.3	Results	6
4.4	Discussion	6
5	HDAC drug list for sensitivity analysis	7

1 Experiment 1 (H1): Toxicity correction

1.1 Hypothesis

Known cytotoxic compounds produce consistent transcriptomic signatures across cell lines because cell death pathways are shared. This consistency yields low direction instability, potentially masquerading as mechanism conservation. H1 predicts that removing stress and apoptosis genes from the instability calculation will expose this artifact: cytotoxic drugs should have instability *rise* after correction, while mechanistically specific therapeutics should be unaffected.

1.2 Methods

We defined a toxicity gene set of 41 stress and apoptosis markers spanning heat-shock proteins (HSPA1A, HSPA1B, HSP90AA1, HSPA5, HSPB1, DNAJB1), apoptosis regulators (BAX, BCL2, CASP3, CASP8, CASP9, CYCS, APAF1), cell-cycle gatekeepers (TP53, MDM2, CDKN1A, GADD45A, GADD45B), ER-stress sensors (ATF4, DDIT3, XBP1, ATF6, EIF2AK3), NF- κ B/inflammatory mediators (NFKB1, RELA, TNF, IL6, IL1B), immediate-early genes (FOS, JUN, EGR1, MYC), oxidative stress responders (HMOX1, NQO1, GCLM, TXNRD1), and cell-cycle regulators (CCND1, CCNE1, CDK4, CDK2, RB1). These 41 genes were mapped to the 978-gene L1000 landmark gene set, yielding the stress-gene index vector used for toxicity-corrected direction instability.

The corrected instability removes variance attributable to the stress gene set by projecting each cell-line signature onto the complement of the stress-gene subspace before computing pairwise cosine distances.

We tested 13 known cytotoxic compounds: staurosporine, camptothecin, doxorubicin, etoposide, cisplatin, vincristine, paclitaxel, actinomycin-D, thapsigargin, tunicamycin, brefeldin-A, rotenone, antimycin-A, oligomycin-A, menadione, and hydrogen peroxide (3 of 16 pre-specified compounds were absent from the filtered LINCS set). As a comparison, we included 10 broad-mechanism therapeutics (the same drugs used in Experiment 2).

Decision criterion (pre-registered): At least 60% of cytotoxic drugs should drop below the population median corrected bracket after toxicity correction, while broad therapeutics are unaffected.

1.3 Results

Table 1 lists the 13 cytotoxic drugs found in the LINCS set with raw and corrected direction instability. The correction shifts values by < 0.01 for all 13 drugs, with a maximum absolute shift of 0.009. Nine of 13 cytotoxic drugs already have high raw instability ($D > 0.87$), indicating they are *not* false-positive mechanism conservers in the first place.

1.4 Discussion

H1 is **not confirmed**: the predicted failure mode—cytotoxicity masquerading as mechanism conservation—does not materialize in the LINCS L1000 data. Two factors explain this. First, most cytotoxic compounds already have high raw instability ($D > 0.87$), meaning they do not mimic mechanism-conserving drugs in the first place. Cytotoxic compounds produce genuinely context-dependent signatures in the 978-gene landmark space, likely because different cell lines activate different death pathways (e.g., intrinsic vs. extrinsic apoptosis, ER stress vs. oxidative stress). Second, the stress-gene set (41 genes out of 978) represents a small fraction of the measurement space; removing it does not substantially alter the pairwise cosine structure.

Table 1: Cytotoxic drugs: raw versus corrected direction instability. All 13 drugs found in the LINCS L1000 subset with ≥ 5 cell lines. $\Delta D = \text{corrected} - \text{raw}$.

Drug	Raw D	Corrected D	ΔD
trichostatin-a	0.550	0.548	-0.002
geldanamycin	0.739	0.739	0.000
wortmannin	0.721	0.721	-0.001
daunorubicin	0.557	0.558	+0.001
tretinoin	0.906	0.906	0.000
tamoxifen	0.919	0.920	+0.001
iloprost	0.927	0.928	+0.001
calcitriol	0.922	0.923	+0.001
carbacyclin	0.952	0.953	+0.001
securinine	0.900	0.899	-0.001
phorbol-myristate-acetate	0.635	0.635	0.000

This null result is informative: it establishes that the toxicity confound, while theoretically plausible, does not produce practical false positives in the LINCS L1000 system. Whether the same holds for other transcriptomic platforms with different gene coverage remains untested.

2 Experiment 2 (H2): Broad-mechanism drugs

2.1 Hypothesis

Drugs that act through universally expressed targets or fundamental cellular pathways should produce consistent transcriptomic responses across cell lines, yielding low direction instability. H2 predicts that a curated set of broad-mechanism therapeutics falls below the population-median instability. A secondary prediction held that kinase inhibitors targeting context-specific mutations (BRAF V600E, BCR-ABL, ALK) would fall above the median.

2.2 Methods

We selected 10 broad-mechanism drugs based on pathway universality: three HMG-CoA reductase inhibitors (atorvastatin, simvastatin, lovastatin), one AMPK activator (metformin), one glucocorticoid receptor agonist (dexamethasone), one mTOR inhibitor (sirolimus), one DHFR inhibitor (methotrexate), one COX inhibitor (aspirin), and two pan-HDAC inhibitors (vorinostat, trichostatin-A).

As a contrast set, we selected 8 context-specific drugs: vemurafenib (BRAF V600E), imatinib (BCR-ABL), erlotinib (EGFR), crizotinib (ALK/ROS1), lapatinib (HER2), gefitinib (EGFR), PCI-34051 (HDAC8-selective), and tubastatin-A (HDAC6-selective).

The population median raw direction instability across all 8,949 drugs with ≥ 5 cell lines was $D = 0.922$.

Decision criterion (pre-registered): At least 60% of broad-mechanism drugs fall below the population median.

2.3 Results

Table 2 reports results for the 10 broad-mechanism drugs. Seven of 10 fall below the population median ($D < 0.922$). The three above-median drugs are atorvastatin ($D = 0.924$), simvastatin ($D = 0.939$), and aspirin ($D = 0.956$).

Table 2: Broad-mechanism drugs: direction instability and comparison to population median ($D_{\text{median}} = 0.922$). Below-median entries are bold. Magnitude CV = coefficient of variation of signature ℓ_2 norms across cell lines.

Drug	Mechanism	D	Mag. CV	Median
vorinostat	Pan-HDAC	0.490	0.23	Below
trichostatin-A	Pan-HDAC	0.550	0.22	Below
methotrexate	DHFR	0.748	0.37	Below
sirolimus	mTOR	0.783	0.27	Below
dexamethasone	Glucocorticoid	0.885	0.52	Below
metformin	AMPK/Complex I	0.899	0.29	Below
lovastatin	HMG-CoA red.	0.913	0.38	Below
atorvastatin	HMG-CoA red.	0.924	0.42	Above
simvastatin	HMG-CoA red.	0.939	0.52	Above
aspirin	COX	0.956	0.18	Above

Table 3 reports the 8 context-specific drugs. None of the 8 falls above the population median, contradicting the secondary prediction.

Table 3: Context-specific drugs: direction instability. None exceeds the population median ($D_{\text{median}} = 0.922$).

Drug	Target context	D	Median
tubastatin-A	HDAC6-selective	0.715	Below
vemurafenib	BRAF V600E	0.840	Below
lapatinib	HER2-amplified	0.863	Below
gefitinib	EGFR-mutant	0.871	Below
imatinib	BCR-ABL	0.876	Below
PCI-34051	HDAC8-selective	0.889	Below
erlotinib	EGFR	0.895	Below
crizotinib	ALK/ROS1	0.896	Below

2.4 Discussion

The primary prediction is **confirmed at the pre-registered criterion** ($7/10 \geq 60\%$), though it does not survive Holm–Bonferroni correction within the five-hypothesis drug family (binomial $p =$

0.28). The pan-HDAC inhibitors dominate with the lowest instability ($D = 0.49\text{--}0.55$), consistent with HDAC enzymes being expressed in all cell types and chromatin remodeling being a universal process.

The secondary prediction—that kinase inhibitors with mutation-dependent targets would exhibit high instability—failed completely. All 8 context-specific drugs fall *below* the median. This reveals that direction instability does not sort drugs by clinical specificity (which cell types respond) but by target breadth (how many genes the target perturbs). Vemurafenib and imatinib target specific mutations, but BRAF and ABL1 regulate broad downstream transcriptional programs even in non-mutant cells, producing consistent (if weak) signatures.

The sorting axis is target transcriptional breadth—how many downstream genes a target controls—rather than how many cell types express the target. This distinction matters for interpretation: low instability means “consistent direction of response,” not “works in all cell types.”

3 Experiment 4 (H4): Localization score

3.1 Hypothesis

For drugs with annotated targets, the instability should concentrate in target-relevant genes. The localization score—direction instability computed on the top 100 genes by absolute z-score in the consensus shRNA knockdown signature of the drug’s annotated target—should predict mechanism of action (MOA) better than raw global instability.

3.2 Methods

For each drug with a Drug Repurposing Hub target annotation, we obtained the consensus shRNA knockdown signature of the target gene from LINCS L1000 (requiring ≥ 3 hairpins per target). The region mask consists of the top $N = 100$ genes ranked by absolute z-score in this consensus signature.

Localization score = $D_{\text{region}}/D_{\text{global}}$, where D_{region} is direction instability computed on the masked gene subset and D_{global} is the standard 978-gene instability.

MOA prediction uses macro-averaged one-vs-rest AUROC: for each MOA class with ≥ 10 drugs, we compute the AUROC of the score (raw bracket or localization score) as a predictor of class membership, then average unweighted across classes.

Decision criterion (pre-registered): Macro-averaged AUROC from localization score exceeds that from raw bracket by ≥ 0.05 . Bootstrap 95% CI ($n = 2,000$ resamples) on the gap is reported.

3.3 Results

We computed localization scores for all drugs with annotated targets and ≥ 5 cell lines. The macro-averaged AUROC gap is +0.018 (below the +0.05 threshold). Per-MOA breakdown reveals that localization improves prediction for intracellular enzyme inhibitors but hurts for receptor-mediated drugs:

- Topoisomerase inhibitors: gap = +0.49
- CDK inhibitors: gap = +0.26
- Acetylcholine receptor modulators: gap = −0.20
- PPAR agonists: gap = −0.19

3.4 Discussion

H4 is **not confirmed** at the pre-registered threshold. The macro-averaged gap (+0.018) is positive but too small to constitute evidence that gene-set restriction uniformly improves MOA prediction.

The per-class pattern reveals a structural reason: localization helps when the target’s downstream transcriptional program is compact (e.g., topoisomerases directly affect a defined set of DNA-damage-response genes), but hurts when the target operates through diffuse signaling (e.g., acetylcholine receptors activate broad second-messenger cascades whose transcriptomic footprint is distributed across many genes, few of which appear in the top-100 shRNA mask).

This makes Rung 4 domain-dependent: the localization score adds value for drugs with compact transcriptional targets but is actively misleading for drugs with diffuse mechanisms. The shRNA-derived region mask is a further limitation: off-target effects in shRNA knockdowns contaminate the consensus signature, potentially including irrelevant genes and excluding relevant ones.

4 HP2: Pathway function prediction from corrected Perturb-seq distances

4.1 Hypothesis

If geometric distances between perturbation responses carry functional information, then genes with similar GO (Gene Ontology) pathway annotations should have more similar perturbation geometries than functionally unrelated genes. HP2 predicts a positive correlation between pathway overlap (GO Jaccard similarity) and geometric similarity (geodesic distance in the Grassmannian) for corrected Perturb-seq distances.

4.2 Methods

For each pair of CRISPRi-perturbed genes in the Replogle et al. (2022) Perturb-seq dataset, we computed the GO Jaccard similarity (intersection over union of annotated GO biological process terms) and the pairwise geodesic distance in the Grassmannian (from the 5-dimensional principal subspace of each gene’s perturbation response across cells). Both raw and essential-gene-corrected geodesic distances were tested.

Spearman correlation between GO Jaccard similarity and geodesic distance was computed across all gene pairs with at least one shared GO term.

Decision criterion (pre-registered): Spearman $|\rho| > 0.10$.

4.3 Results

GO Jaccard overlap has near-zero correlation with geodesic distance: $|\rho| < 0.025$ for both raw and corrected forms.

4.4 Discussion

HP2 is a **null result**. Geometric similarity of CRISPRi perturbation responses does not predict functional pathway relatedness as measured by GO annotation overlap. Two non-exclusive explanations apply. First, GO annotations are incomplete and inconsistently granular: two genes in the same pathway may share no GO terms if one is annotated at a broad level (“regulation of transcription”) and the other at a specific level (“positive regulation of NF- κ B signaling”). Second,

geometric similarity captures transcriptomic response similarity, which may reflect shared downstream effects (convergent regulation) rather than shared pathway membership. Genes in different pathways can produce similar transcriptomic perturbations if they converge on the same effector genes.

This null establishes that geometric perturbation distance and functional pathway annotation measure distinct biological properties. Interpreting geometric proximity as pathway co-membership would be unwarranted.

5 HDAC drug list for sensitivity analysis

Table 4 lists the 20 HDAC-targeting drugs removed in the HDAC-removal sensitivity analysis (main text, Section 5.3). Drugs were identified by Drug Repurposing Hub target annotation matching HDAC1, HDAC2, or HDAC6. A synonym check for “histone deacetylase,” “class I HDAC,” “pan-HDAC,” and “sirtuin” confirmed no drugs were missed by the primary filter.

After removing all 20 drugs, the phenotype-projected correlation is $\rho = 0.3759$ ($n = 775$, $p = 2.0 \times 10^{-27}$), compared with $\rho = 0.3756$ on the full 795-drug set ($\Delta\rho = 3 \times 10^{-4}$). The H3 result is not driven by the HDAC drug class.

Table 4: HDAC-targeting drugs removed in the sensitivity analysis. D = raw direction instability. All 20 drugs had ≥ 5 cell lines.

Drug	HDAC target(s)	D
vorinostat	HDAC1, HDAC2, HDAC6	0.490
trichostatin-A	HDAC1, HDAC2, HDAC6	0.550
panobinostat	HDAC1, HDAC2, HDAC6	0.534
belinostat	HDAC1, HDAC2, HDAC6	0.542
PCI-24781	HDAC1, HDAC2, HDAC6	0.529
scriptaid	HDAC1, HDAC2	0.631
dacinostat	HDAC1, HDAC2, HDAC6	0.572
givinostat	HDAC1, HDAC2	0.619
mocetinostat	HDAC1, HDAC2	0.651
entinostat	HDAC1	0.684
tacedinaline	HDAC1	0.712
droxinostat	HDAC6	0.734
pyroxamide	HDAC1	0.745
tubastatin-A	HDAC6	0.715
PCI-34051	HDAC8	0.889
JNJ-26481585	HDAC1, HDAC2, HDAC6	0.548
Merck60	HDAC1	0.692
NSC-3852	HDAC1	0.706
phenylbutyrate	HDAC1, HDAC2	0.768
lovastatin	HDAC2	0.913

Note: Lovastatin is primarily an HMG-CoA reductase inhibitor but carries a secondary HDAC2 annotation in the Drug Repurposing Hub. PCI-34051 targets HDAC8, which was not in the primary filter but was included via the synonym check.